

Summer Placement Self-Assessment 1

Precalculus (SPreCalc8M) · Complete Step-by-Step Solutions Reference · 44 / 50 pts (88%)

Q1 | Exponent Rules

SPreCalc8M 1.2.027 Score: 2/2

Simplify each expression and eliminate any negative exponents.

(a) x^2y^{-2} / x^{-8}

Divide powers of the same base by subtracting exponents:

$$x^2 / x^{-8} = x^{2-(-8)} = x^{10}$$

y^{-2} has a negative exponent — move it to the denominator as y^2 .

Answer: x^{10} / y^2

(b) $(a^3 / 2b^3)^3$

Apply the exponent to every factor in numerator and denominator:

$$(a^3)^3 / (2b^3)^3 = a^9 / 2^3b^9$$

Simplify $2^3 = 8$.

Answer: $a^9 / 8b^9$

Q2 | Number Classification

SPreCalc8M 1.1.001 Score: 4/4

Give an example of each of the following.

Type	Example	Explanation
(a) Natural number	2	Counting numbers {1, 2, 3, ...}
(b) Integer, not natural	-4	Integers include negatives and zero
(c) Rational, not integer	1/2	Ratio p/q (q≠0) that doesn't reduce to a whole number
(d) Irrational number	π	Non-terminating, non-repeating decimal; cannot be written as p/q

Q3 | Polynomial Subtraction

SPreCalc8M 1.3.018 Score: 1/1

Simplify: $(8x^2 + x + 1) - (7x^2 - 8x - 15)$

Step 1 – Distribute the negative sign across the second parentheses:

$$8x^2 + x + 1 - 7x^2 + 8x + 15$$

Step 2 – Group like terms:

$$(8x^2 - 7x^2) + (x + 8x) + (1 + 15)$$

Answer: $x^2 + 9x + 16$

Q4 | Rational Expression Subtraction

SPreCalc8M 1.4.042 Score: 1/1

Simplify: $(8x - 1)/(x + 4) - 2$

Step 1 – Rewrite 2 with the same denominator: $2 = 2(x+4)/(x+4)$

Step 2 – Subtract:

$$[8x - 1 - 2(x+4)] / (x+4) = [8x - 1 - 2x - 8] / (x+4) = (6x - 9)/(x+4)$$

Answer: $(6x - 9) / (x + 4)$

Q5 | Factoring Quadratics

SPreCalc8M 1.5.507.XP Score: 1/1

Solve by factoring: $x^2 + 2x - 3 = 0$

Step 1 – Find two numbers that multiply to -3 and add to 2 : those are 3 and -1 .

Step 2 – Factor: $(x + 3)(x - 1) = 0$

Step 3 – Set each factor to zero:

$$x + 3 = 0 \rightarrow x = -3$$

$$x - 1 = 0 \rightarrow x = 1$$

Answer: $x = -3, 1$

Q6 | Modeling with Variables

SPreCalc8M 1.7.014 Score: 1/1

Total rent paid at \$805/month; n = number of months.

Total cost = (rate per month) $\times$ (number of months)

Answer: $\$805n$

Q7 | Absolute Value Inequalities

SPreCalc8M 1.8.079 Score: 2/2

Solve: $|3x + 4| < 5$

Step 1 – For $|A| < c$ (with $c > 0$), rewrite as a double inequality:

$$-5 < 3x + 4 < 5$$

Step 2 – Subtract 4 from all three parts:

$$-9 < 3x < 1$$

Step 3 – Divide all parts by 3:

$$-3 < x < 1/3$$

Answer: Interval notation: $(-3, 1/3)$

Graph: Open circles at $x = -3$ and $x = 1/3$ with the segment between them shaded.

Q8 | Distance & Midpoint

SPreCalc8M 1.9.025 Score: 3/3

Points: $(0, 8)$ and $(8, 14)$

(a) Plot the points — place $(0, 8)$ and $(8, 14)$ on the coordinate grid.

(b) Distance:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(8 - 0)^2 + (14 - 8)^2} = \sqrt{64 + 36} = \sqrt{100}$$

Answer: $d = 10$

(c) Midpoint:

$$\text{Midpoint} = ((x_1 + x_2)/2, (y_1 + y_2)/2)$$

$$= ((0 + 8)/2, (8 + 14)/2) = (8/2, 22/2) = (4, 11)$$

Answer: $(x, y) = (4, 11)$

Q9 | Linear Functions from Graphs

SPreCalc8M 2.5.034 Score: 1/2

The graph of linear function f is given.

(a) Rate of change (slope):

Read two clear points from the line and compute $\Delta y / \Delta x$. The line falls steeply.

Answer: Slope = -4 ✓

(b) Express f in the form $f(x) = ax + b$:

We know $a = -4$. Find b by substituting a known point on the line. If the line crosses the x -axis at $x = 3/4$:

$$0 = -4(3/4) + b \rightarrow 0 = -3 + b \rightarrow b = 3$$

Answer: $f(x) = -4x + 3$

■ The submitted answer $f(x) = -4x + 2$ was marked INCORRECT. The slope is right — only the y -intercept needs to be re-read carefully from your specific graph. Always verify by plugging the x -intercept back in to solve for b .

Q10 | Domain & Range of Square Root

SPreCalc8M 2.3.024 Score: 2/2

$$f(x) = \sqrt{x - 5}$$

Domain: The radicand must be ≥ 0 :

$$x - 5 \geq 0 \rightarrow x \geq 5$$

Answer: Domain: $[5, \infty)$

Range: A square root always outputs values ≥ 0 ; as $x \rightarrow \infty$, $f(x) \rightarrow \infty$.

Answer: Range: $[0, \infty)$

Q11 | Table of Values & Graphing

SPreCalc8M 2.2.518.XP Score: 7/7

$f(x) = x^2 - 1$ — substitute each x value:

x	Calculation	f(x)
± 5	$(\pm 5)^2 - 1 = 25 - 1$	24
± 4	$16 - 1$	15
± 3	$9 - 1$	8
± 2	$4 - 1$	3
± 1	$1 - 1$	0
0	$0 - 1$	-1

Graph: This is $y = x^2$ shifted **down 1 unit**. Upward-opening parabola, vertex at $(0, -1)$, passing through $(\pm 1, 0)$.

Q12 | Net Change & Average Rate of Change

SPreCalc8M 2.4.510.XP Score: 1/2

General method for $h(t)$ evaluated between $t = a$ and $t = b$:

(a) Net change:

$$\text{Net change} = h(b) - h(a)$$

Positive result $\rightarrow$ function increased; negative $\rightarrow$ decreased.

(b) Average rate of change:

$$[h(b) - h(a)] / (b - a)$$

This equals the slope of the secant line connecting $(a, h(a))$ and $(b, h(b))$.

■ A common error (the one that cost 1 point here) is dropping or mis-tracking a negative sign when subtracting $h(a)$, which can flip the sign of the final answer. Always substitute carefully and check the sign of the result.

Q13 | Function Transformations

SPreCalc8M 2.6.032 Score: 1/1

Sketch $f(x) = (x + 2)^2$ using transformations.

Step 1 – Start with the parent graph $y = x^2$ (upward parabola, vertex at origin).

Step 2 – Replacing x with $(x + 2)$ shifts the graph **horizontally left by 2 units**.

Rule: $y = f(x + h)$ shifts the graph left by h (when $h > 0$).

Step 3 – New vertex: $(0, 0) \rightarrow (-2, 0)$.

Answer: Graph: Standard upward parabola shifted 2 units LEFT, vertex at $(-2, 0)$.

Q14 | Function Composition

SPreCalc8M 2.7.514.XP Score: 1/2

$$f(x) = 4x - 2, g(x) = 5 - x^2$$

(a) $(f \circ f)(-1) = f(f(-1))$:

Inner: $f(-1) = 4(-1) - 2 = -4 - 2 = -6$

Outer: $f(-6) = 4(-6) - 2 = -24 - 2 = -26$

Answer: $(f \circ f)(-1) = -26$ ✓

(b) $(g \circ g)(-1) = g(g(-1))$:

Inner: $g(-1) = 5 - (-1)^2 = 5 - 1 = 4$

Outer: $g(4) = 5 - (4)^2 = 5 - 16 = -11$

Answer: $(g \circ g)(-1) = -11$

■ Submitted answer -29 was INCORRECT. The correct value is -11 . Common mistake: computing $5 - (5-1)^2$ instead of first evaluating $g(-1) = 4$, then substituting 4 into g . Always fully evaluate the inner function before applying the outer.

Q15 | Inverse Functions from a Table

SPreCalc8M 2.8.033.MI Score: 1/1

One-to-one function table:

x	1	2	3	4	5	6
$f(x)$	3	7	5	6	9	0

Find $f^{-1}(f(6))$.

By the definition of an inverse: $f^{-1}(f(x)) = x$ for all x in the domain.

Therefore $f^{-1}(f(6)) = 6$ immediately.

Verification: $f(6) = 0$, so $f^{-1}(f(6)) = f^{-1}(0)$. From the table, $f(6)=0$ means $f^{-1}(0)=6$. ✓

Answer: $f^{-1}(f(6)) = 6$

Q16 | Local Extrema of Polynomials

SPreCalc8M 3.2.064 Score: 2/2

$$y = 2x^3 + 11x$$

Consider the derivative: $y' = 6x^2 + 11$.

Since $6x^2 \geq 0$ for all real x , we have $y' = 6x^2 + 11 \geq 11 > 0$ everywhere.

A positive derivative everywhere means the function is strictly increasing — it never turns around.

Answer: No local maximums and no local minimums.

Q17 | Polynomial Long/Synthetic Division

SPreCalc8M 3.3.520.XP Score: 0/1

$$P(x) = x^4 + 2x^3 - 11x, D(x) = x - 3$$

Use synthetic division with root 3. Write all coefficients (include 0 for missing x^2 and constant terms):

Coefficients: 1 2 0 -11 0

Operation	1	2	0	-11	0
Bring down	1				
×3, add		5			
×3, add			15		
×3, add				34	
×3, add					102

Quotient: $Q(x) = x^3 + 5x^2 + 15x + 34$, Remainder: $R = 102$

Answer: $P(x) = (x - 3)(x^3 + 5x^2 + 15x + 34) + 102$

Check: Expand and add 102 $\rightarrow x^4 + 2x^3 - 11x \checkmark$

■ The submitted answer was incorrect. Be sure to include all missing-degree terms (as zeros) in the coefficient row before performing synthetic division.

Q18 | Polynomial Inequalities

SPreCalc8M 3.7.008 Score: 0/1

$$\text{Solve: } 2x^3 - 32x < x^2 - 16$$

Step 1 – Move everything to one side:

$$2x^3 - x^2 - 32x + 16 < 0$$

Step 2 – Test $x = 4$: $2(64) - 16 - 128 + 16 = 0 \rightarrow (x - 4)$ is a factor.

Step 3 – Synthetic division by 4 gives quotient $2x^2 + 7x - 4$.

Step 4 – Factor the quadratic: $2x^2 + 7x - 4 = (2x - 1)(x + 4)$

Full factorization: $(x - 4)(2x - 1)(x + 4) < 0$

Step 5 – Critical points: $x = -4, 1/2, 4$. Sign analysis:

Interval	Test pt	$(x-4)$	$(2x-1)$	$(x+4)$	Product	Sign
$(-\infty, -4)$	$x=-5$	–	–	–	–	– ✓
$(-4, 1/2)$	$x=0$	–	–	+	+	+

(1/2, 4)	x=1	-	+	+	-	- ✓
(4, ∞)	x=5	+	+	+	+	+

Answer: $(-\infty, -4) \cup (1/2, 4)$

Q19 | Exponential Transformations

SPreCalc8M 4.1.510.XP Score: 4/4

$$g(x) = 2^x - 2$$

Step 1 – Start from parent graph $y = 2^x$ (increasing exponential, asymptote $y = 0$, through (0,1)).

Step 2 – Subtracting 2 shifts the entire graph DOWN 2 units.

Step 3 – Horizontal asymptote moves from $y = 0$ to $y = -2$.

Key point check: $(0, 1) \rightarrow (0, -1) \mid (1, 2) \rightarrow (1, 0) \mid (2, 4) \rightarrow (2, 2)$

Answer: Graph: $y = 2^x$ shifted down 2 units (upper-left graph option, rising curve with asymptote $y = -2$)

Property	Answer
Domain	$(-\infty, \infty)$
Range	$(-2, \infty)$ — since $2^x > 0$ always, $2^x - 2 > -2$
Asymptote	$y = -2$

Q20 | Logarithm $\leftrightarrow$ Exponential Form

SPreCalc8M 4.3.504.XP Score: 2/2

Convert to exponential form using: $\ln(A) = B \blacksquare A = e^B$

(a) $\ln(x + 7) = 3$

Answer: $x + 7 = e^3$

(b) $\ln(x - 9) = 5$

Answer: $x - 9 = e^5$

Q21 | Laws of Logarithms

SPreCalc8M 4.4.011 Score: 1/1

Evaluate: $\log_2(40) - \log_2(10)$

Apply the Quotient Law: $\log_b(A) - \log_b(B) = \log_b(A/B)$

$$\log_2(40) - \log_2(10) = \log_2(40/10) = \log_2(4)$$

Since $2^2 = 4$, we have $\log_2(4) = 2$.

Answer: 2

Q22 | Inverse Trigonometry

SPreCalc8M 5.5.507.XP Score: 1/1

Find: $\tan(\sin^{-1}(\sqrt{2}/2))$

Step 1 – Let $\theta = \sin^{-1}(\sqrt{2}/2)$. By definition, $\theta \in [-\pi/2, \pi/2]$ and $\sin \theta = \sqrt{2}/2$.

Step 2 – Recognize the angle: $\sin(\pi/4) = \sqrt{2}/2 \rightarrow \theta = \pi/4$ (45°).

Step 3 – Evaluate: $\tan(\pi/4) = \sin(\pi/4) / \cos(\pi/4) = (\sqrt{2}/2) / (\sqrt{2}/2) = 1$.

Answer: 1

Q23 | Sine Function Transformations

SPreCalc8M 5.3.007 Score: 3/3

$f(x) = -3 \sin(x)$

Step 1 – Start with parent $y = \sin(x)$, oscillating between -1 and 1 .

Step 2 – The factor of 3 stretches the amplitude: now oscillates between -3 and 3 .

Step 3 – The negative sign reflects the graph over the x-axis (starts by dipping downward).

Answer: Graph: Reflected & vertically stretched sine wave, amplitude 3, starting downward at $x = 0$

Property	Answer
Domain	$(-\infty, \infty)$
Range	$[-3, 3]$ — since $\sin x \in [-1, 1]$, multiplying by -3 gives $[-3, 3]$
Amplitude	3
Period	2π (unchanged)

Q24 | Period of Cotangent

SPreCalc8M 5.4.026.MI Score: 1/2

$y = 5 \cot(\pi/2 \cdot x)$

Period formula for $y = A \cdot \cot(Bx)$:

$Period = \pi / |B|$

(Note: cotangent's base period is π , NOT 2π like sine/cosine.)

Here $B = \pi/2$, so:

$Period = \pi / |\pi/2| = \pi \times (2/\pi) = 2$

Answer: Period = 2

■ Submitted answer $\pi/2$ was **INCORRECT**. A common error is using the formula $2\pi/|B|$ (which applies to sine & cosine) instead of the cotangent formula $\pi/|B|$. The factor of 5 affects the vertical stretch only — it does not change the period.

Graph notes: Vertical asymptotes occur wherever $\pi/2 \cdot x$ is a multiple of π , i.e. at every even integer $x = \dots, -2, 0, 2, 4, \dots$. The branches between asymptotes are spaced 2 units apart.

Q25 | Vector Representation

SPreCalc8M 8.5.025 Score: 1/1

Sketch vector $u = \langle -3, 4 \rangle$ with initial points $(0,0)$, $(2,3)$, and $(-3,5)$.

A vector $\langle -3, 4 \rangle$ means: from any initial point, move 3 left and 4 up to reach the terminal point.

Initial Point	Calculation	Terminal Point
$(0, 0)$	$(0+(-3), 0+4)$	$(-3, 4)$
$(2, 3)$	$(2+(-3), 3+4)$	$(-1, 7)$
$(-3, 5)$	$(-3+(-3), 5+4)$	$(-6, 9)$

Magnitude: $|u| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

Direction: slope = $\Delta y / \Delta x = 4 / (-3)$ — the arrow points up and to the left.

Answer: Three identical arrows of length 5, all pointing in the same up-left direction, starting from each initial point.

Quick Reference Summary

Q	Topic	Key Takeaway
Q 1	Exponent Rules	Subtract exponents dividing same base; distribute outer exponent over quotient
Q 2	Number Classification	Natural $\subset$ Integer $\subset$ Rational $\subset$ Real; Irrational $\not\subset$ Rational
Q 3	Polynomial Subtraction	Distribute the negative sign carefully before combining like terms
Q 4	Rational Expr. Subtraction	Get a common denominator before subtracting
Q 5	Factoring Quadratics	Find two numbers multiplying to c and summing to b
Q 6	Modeling with Variables	Total = rate $\times$ quantity
Q 7	Absolute Value Inequalities	$ A < c \Rightarrow -c < A < c$
Q 8	Distance & Midpoint	$d = \sqrt{(\Delta x^2 + \Delta y^2)}$; midpoint = average the coordinates
Q 9	Linear Functions (Graphs)	Slope correct (-4); re-read y-intercept b carefully from the graph
Q 10	Domain/Range of $\sqrt{\quad}$	Radicand ≥ 0 for domain; output ≥ 0 for range
Q 11	Table & Graph $x^2 - 1$	Vertical shift down 1 unit from parent $y = x^2$
Q 12	Net Change / Avg Rate	$h(b) - h(a)$ and $[h(b) - h(a)] / (b - a)$ — watch signs carefully
Q 13	Transformations	$(x + 2)^2$ shifts graph LEFT 2 units
Q 14	Function Composition	$(g \circ f)(-1) = -11$, not -29; evaluate inside-out

Q 1 5	Inverse Functions	$f^{-1}(f(x)) = x$ always, for any one-to-one function
Q 1 6	Local Extrema	Derivative always positive $\Rightarrow$ always increasing $\Rightarrow$ no extrema
Q 1 7	Synthetic Division	Correct: $Q(x)=x^3+5x^2+15x+34$, $R=102$; include zero coefficients
Q 1 8	Polynomial Inequalities	Fully factor, do sign analysis: $(-\infty, -4) \cup (1/2, 4)$
Q 1 9	Exponential Transformations	Vertical shift moves the asymptote
Q 2 0	Log $\leftrightarrow$ Exponential Form	$\ln A = B \quad \blacksquare \quad A = e^B$
Q 2 1	Laws of Logarithms	$\log_b(A) - \log_b(B) = \log_b(A/B)$
Q 2 2	Inverse Trig	$\sin^{-1}(\sqrt{2}/2) = \pi/4$; then $\tan(\pi/4) = 1$
Q 2 3	Sine Transformations	Negative $\Rightarrow$ reflect; coefficient $\Rightarrow$ amplitude
Q 2 4	Period of Cotangent	Period = $\pi/ B $, NOT $2\pi/ B $ (cotangent base period is π)
Q 2 5	Vector Translation	Add the vector's components to each initial point to get terminal point